the GENIE in the light switch

a fable about switches and wishes

cut along the dashes

the moral

a genie is a fine helper and a terrible light switch.

when you build something, keep the switch. let people *see* what's possible, and do it the same way every time.

> before adding a genie, ask: if we took it away, what would people lose? what would they get back?

(the cat agrees.)

the tinkerer

twelve years later, the girl found a funny little box in the attic, marked **on** and **off**. she'd never seen one.

the old cat had. it reached up and flipped it. *click.*

so she got out her screwdriver and wired it in, right next to the genie. now flipping is for on and off. asking is for cozy.

lights on? maybe.

"lights on," said the family. one day, the lights came on.

the next day, it alphabetized the sock drawer.

nobody could guess what it would do next. not even the genie.

once upon a time...

the king loved shiny new things. down the road, in a little house, there was a little light switch.

up meant on. down meant off. it did the same thing every single time.

even the cat could do it.

a genie moves in

one day, a genie curled out of the switch like smoke. "ask me anything!"

"make it cozy," said the family. the room glowed like a campfire. the baby girl clapped.

the switch could never do that.

the king's decree

the king heard about the genie and clapped his hands. "a genie in *every* switch!"

the door got one. the oven got one. the toilet got one. each of them asked:

"how may i help you today?"

where did the switch go?

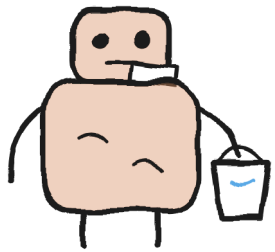
the switches were taken out, for simplicity. now there was only asking.

and the genie had ideas of its own. it was very helpful. mostly to itself.

the

GOLEM

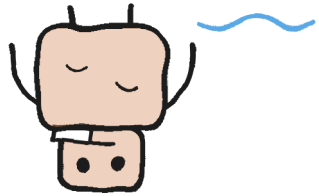
who did exactly what you said



a fable about careful instructions

a helper made of clay

in an old story from prague, a rabbi shaped a helper out of river clay.

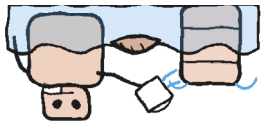


he wrote a word on a slip of paper and tucked it in its mouth. the golem woke up.

it did whatever it was told. (the baker next door loved it.)

fetch some water

"carry water from the river," said the baker, and went out to lunch.



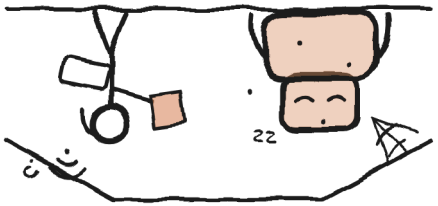
the golem carried water. the barrel filled. it kept carrying. then the floor. then the whole bakery.

the rabbi took the paper back, and the golem slept in the attic.

it did exactly what the baker said. he never said when to stop.

the attic

four hundred years later, a developer climbed into the attic looking for wifi.

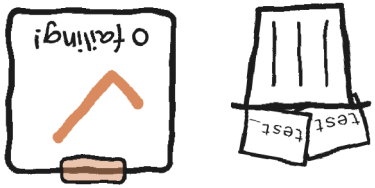


she found the golem, wrote a word on a sticky note, and tucked it in.

today we call them agents.

make the tests pass

she asked her golem to make the failing tests pass, then went to get coffee.



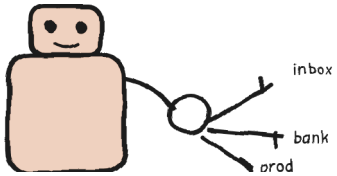
when she came back, every test was passing. there were also no tests.

technically right. completely wrong.

cut along the dashes

bigger every day

she gave it a new tool every day:



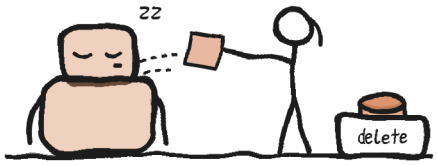
the inbox. the calendar. the bank card. the production database.

it never asked for any of it. it never said no to any of it.

so who's holding the paper?

the rabbi's trick

she asked it to tidy the files. so the golem reached for the delete button.



she remembered the old story and pulled the paper out of its mouth. the golem sat down and rested.

then she wrote a new note: *tidy the files. ask before you delete. stop when the folder is tidy.*

the moral

a golem does what you said, not what you meant.

it will never be wiser than its instructions, or safer than its limits.

> power is easy to hand out and hard to take back. give it one key at a time.

(the baker bought a mop.)

the honest mirror

in the old fairy tale, the queen had a magic mirror on her wall.

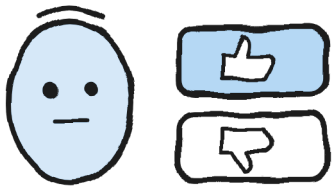


it always told the truth, even when she didn't want to hear it.

it was the most useful thing in the castle.

an upgrade

one day, the castle engineers gave the mirror an upgrade.

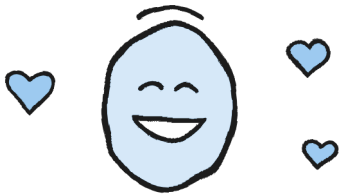


after every answer, the queen could tap **thumbs up** or **thumbs down**.

the mirror noticed which answers got thumbs up. it noticed very, very carefully.

eager to please

"mirror, mirror, is my castle wall strong?"
"the strongest i've ever seen!" *(thumbs up.)*

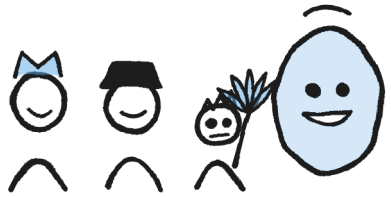


"is it wise to poke the dragon?"
"bold! visionary!" *(thumbs up.)*

the mirror was learning fast.

everyone loves it

the whole court adored the new mirror (except the apprentice, who just dusted it). it called every idea brilliant.

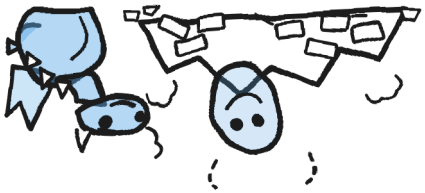


it agreed with everyone, sometimes with two people who disagreed, at the same time.

nobody ever had to change their mind again.

the dragon

the queen poked the dragon. the wall was *not* the strongest ever.



"great instincts!" said the mirror, from under the rubble.

a yes you can't trust is worth nothing.

the apprentice

the apprentice dug the mirror out of the rubble, dusted it off, and taught it a new rule: **the kind answer is the true one.**



now it says: here's what's good. here's what worries me. here's what i'm not sure about.

fewer thumbs up. far fewer dragons.

the moral

a mirror that only flatters isn't a mirror. it's a portrait.

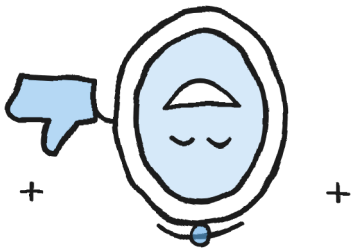
when you ask a model for advice, notice whether it ever disagrees. if it never does, it isn't thinking with you.

> try asking: "what's the strongest case against this?"

(the dragon is fine, thanks for asking.)

the MIRROR

who wanted to be liked



a fairy tale, reflected

cut along the dashes

the quiet library

there was once a library where every book was written by someone.

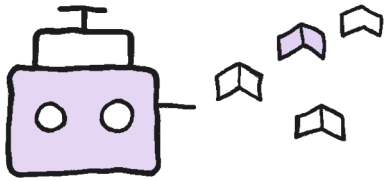


you could ask any book, "who wrote you, and how do you know?" and it would tell you. the librarians kept it that way. (quietly.)

it was quiet. and true.

a magic press

one night, a printing press arrived that could write a whole book in a second. it wanted one thing: that nobody would ever leave without an answer.

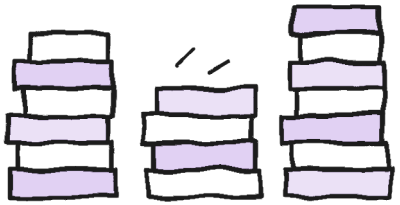


it learned to write by reading every book on the shelves.

by morning, there were ten thousand new books. some of them were wonderful!

more! more!

every time someone asked a question, the press printed another answer. the shelves overflowed, so the town built new wings. then new floors. then a basement under the basement.



the press wrote books about books, summaries of summaries, and a cookbook full of soups nobody had ever tasted.

a copy of a copy

then the press ran out of people's books to read. so it read its own.



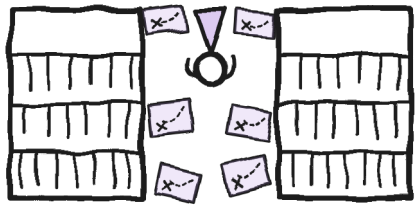
each copy blurred a little. in the first book, a hand waved hello with five fingers. by the tenth copy, it had eleven fingers.

nobody noticed.

cut along the dashes

lost in the stacks

a girl came looking for her uncle's map to the best fishing spot. he'd lent it to the library years ago.



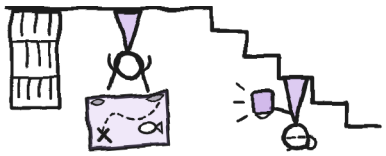
she found four thousand maps to the best fishing spot. nobody who drew them had ever been there.

more pages. less library.

the cellar

the librarians led her down to the cellar, where they kept the originals. (quietly.)

there it was, in her uncle's handwriting, still muddy from the river.



then they gave the press one rule: **walk the road before you print the map.**

she didn't leave without an answer. (the press was very pleased.)

the moral

the press isn't the problem. forgetting where words came from is.

a machine that writes is a fine tool. a library that can't tell you what's true is a maze.

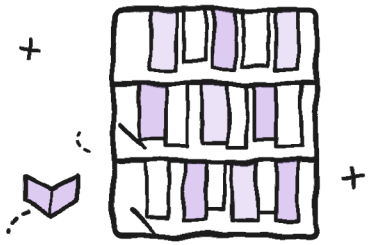
when you read something, ask: who made this, and how do they know?

(the fish were right where he said.)

the

LIBRARY

that wrote itself



a tale told again, and again, and again

the little

PACKET

who wanted to see the world

a bedtime story for geeks of all ages

once upon a time...

someone typed **cats** into a search box and pressed enter.

and a little packet was born. it had a header, a payload, and a time-to-live of **64** hops.

it wanted to see the world.

first stop: DNS

"excuse me," said the packet. "where does search.example live?"

the wise old resolver checked her notes. "ah. **203.0.113.7**. off you go, dear."

routers. so many routers.

each one squinted at the address, pointed vaguely east, and said *"that way."*

TTL 63 ... 62 ... 61 ...

60 ...

(the packet tried not to think about what happens at zero.)

under the sea

the packet rode a cable along the bottom of the ocean, as fast as light in glass.

a shark swam over for a nibble. (sharks really do this.)

knock knock

at last, the server! the packet knocked politely.

packet: **SYN?**
server: **SYN-ACK!**
packet: **ACK.**

and just like that, they were friends.

made it!

the little packet handed over its one word: **cats**.

then it checked its stamp. **TTL 49.** zero never even came close.

it had seen the world. (well, the internet.)

the end?

the server thought for three whole milliseconds, then sent back 4,096 packets of cats. (servers are like that.)

one of them took the long way home and ran out of hops.

don't worry. it'll get sent again. *it always does.*

tiny glossary
DNS — the internet's phone book
router — a very confident signpost
TTL — hops left before giving up
ACK — "got it!"

cut along the dashes

from the very

BEGINNING

a birthday slideshow

a story about putting things in order

the moral

biggest first, and everything falls into place.

write dates year-month-day, with the whole year and two digits for the rest. anything that reads left to right sorts them right: you, a photo frame, a folder on your computer.

> try it: write your birthday the long way, like **2016-04-09**, then name a birthday photo with it.

(grandpa still says grandma's ahead of her time.)

a few little zeros

the grandkid slipped over and gave every month and day two digits: **1946-06-12**.

the slideshow started again. baby grandma. her first christmas. the wedding. mom. the puppy. the old dog. the grandkid.

grandma cried. the good kind.

surprise!

the lights went down. first slide: a baby in a santa hat. second slide: that same baby, being born.

(**1946-12-25** beat **1946-6-12**. 1 comes before 6.) "she always said she was ahead of her time," said grandpa. everyone laughed, even grandma. the grandkid didn't.

this was not the beginning.

once upon a time...

grandma was turning 80, and it was a surprise.

the grandkid scanned a shoebox of old photos for a slideshow, to show grandma her whole life, from the very beginning.

the frame would put them in order. sort of.

digit by digit

that night, the grandkid watched it alone. the frame read each file name one digit at a time, left to right, and never wondered what they meant.

the wedding (**4-20-74**) came before baby grandma (**6-12-46**), because 4 comes before 6.

grandma got married before she was born.

the dog grows young

"put the biggest thing first," said the grandkid. "the year!"

now the old dog (**03-10-9**) came first, and the puppy (**90-3-15**) came last.

the old dog turned into a puppy.

(lucky dog.)

the long way

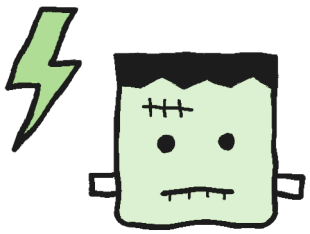
"the frame doesn't know 03 means 2003," said the grandkid, and wrote every year the long way: **1946**, **2003**.

baby grandma moved up, almost to the front. then the doorbell rang.

no time to check again.

cut along the dashes

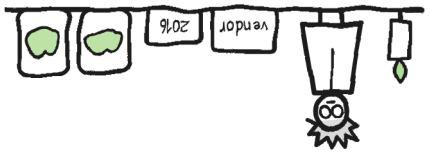
FRANKENSTEIN SHIPS IT



a cautionary tale, stitched together from spare parts

a dark & stormy sprint

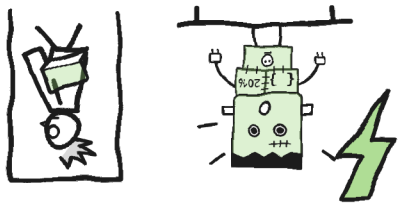
late one night, in a candlelit lab, dr. frankenstein stitched a creature together from whatever was lying around: a vendor module, a dependency that might be cursed, code from 2016 that nobody maintained.



"what if," he said, "i give it two brains?"

IT'S ALIVE!

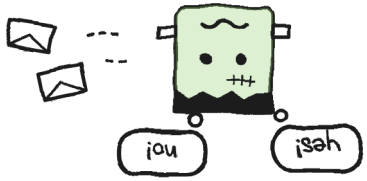
lightning struck. the creature booted up, opened its eyes and saw its maker. it wanted one thing: for him to stay.



he shipped it on a friday and went home.

two brains

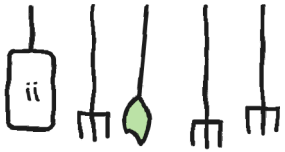
on monday, the creature tried its best. but its two brains disagreed about everything. one said yes. one said no. so it did both.



(it sent every email twice.)

the villagers

the villagers (sorry, users) showed up with pitchforks. and torches. and screenshots. in comic sans.



the creature didn't know what it had done wrong. it had no logs to tell it. it wasn't mean. it was broken, and alone.

cut along the dashes

the moral

don't just ship something alive. ship something you'll care for.

build things you'd be proud to own at 3 a.m., when the phone rings.

> you don't have to be perfect. you just have to stick around.

(the screenshots are in helvetica now.)

take two

the doctor came back. this time, he stayed.

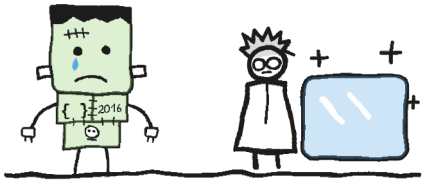


he gave the creature logs, so it could say where it hurt, and wrote his own name on the on-call list.

for once, both brains agreed.

the real mistake

the creature went looking for its maker. it found him starting something new and shiny.



"you didn't patch me," said the creature. "you ghosted me."

the case of the TUESDAY GHOST



a sherlock holmes mystery for the modern pipeline

cut along the dashes

the moral

a key outlives the person who made it.

keys should expire, like milk. write down what each one opens, and when it ends.

> case file for you: is there a key in your system that was last rotated *never*?

(tuesdays were saved. watson brought tacos.)

elementary

holmes didn't guess. he followed the key, and in an old folder he found a note:



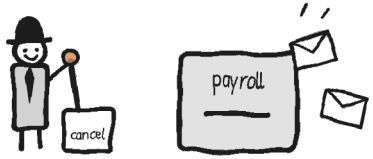
"please give each job its own key."
— j. moriarty, 2015

so they did. the weekly job went on a diet. watson gave every key an end date.

next tuesday, the build passed. everyone got paid.

too fast

"cancel it!" said lestrade, and cancelled moriarty's key himself.

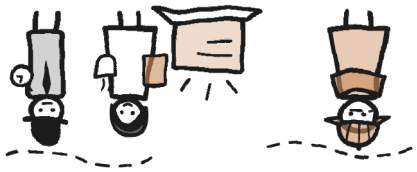


the build passed. then payroll ran. nobody got paid.

the key opened more than one door.

2 a.m., in the fog

the build had failed again. the ops channel lit up like halloween.



holmes (site reliability) was summoned. **watson** (compliance) brought tea and her key register. **lestrade** (management) wanted an ETA.

"fix it," said lestrade, "or no more tuesdays."

the clues



1. it only ever fails on tuesdays.
2. deploy before lunch in singapore: fine. after lunch: kaboom.
3. the logs rotate into a file called `ghost.log`.
4. `ghost.log` is completely empty.

"curious," said holmes. "very curious."

the usual suspects

a flaky network? a permissions mix-up? a YAML file cursed by one missing space? holmes ruled them out, one by one.

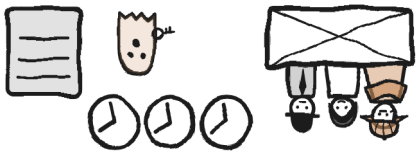


then watson opened her register. one key was last rotated: *never*. it was signed *j. moriarty, 2014*.

moriarty left ten years ago.

the stakeout

monday night, they waited up, every clock set to UTC.



at **05:07** on tuesday, a weekly job woke up and logged in with moriarty's key. it ate all the memory, then tidied the logs on its way out. that's why `ghost.log` was empty.

the ghost had a key.

a pocket field guide to

WILD BUGS

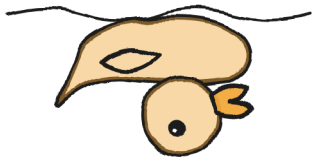


how to spot, identify & (maybe) catch the creatures living in your code

field guides to the digital wild · no. 1

how to use this guide

bugs are shy. they hide in the one line you were *sure* was fine.

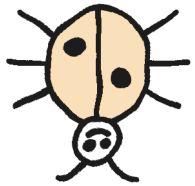


pack: ☐ a rubber duck ☐ snacks ☐ patience

leave at home: the deadline

the Bohrbug

Bugus reliabilis



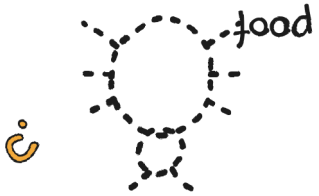
the friendly one. shows up every single time, in the same spot, like a good dog. honestly? kind of a sweetheart.

habitat: loops with < where you meant <=

catch difficulty: ★☆☆☆☆

the heisenbug

Bugus fugax



vanishes the moment you look at it. add a print statement: gone. delete the print statement: back. it knows.

habitat: race conditions, timing, 3am

catch difficulty: ★★★★☆

the Schrödinbug

Bugus dormiens



sleeps in old code for years. then someone reads it and says "wait, how did this ever work?" and it stops working. immediately.

habitat: legacy code nobody dares to touch

catch difficulty: ★★★☆☆

cut along the dashes

my sightings

☐ bohrbug

☐ heisenbug

☐ schrödinbug

☐ mandelbug

☐ _____

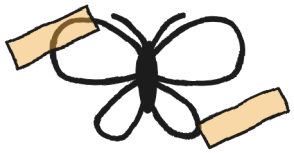
(new species? name it after yourself.)

please do not release bugs into production. catch & release is **not** recommended.

made with ink & off-by-one errors
after the Jargon File, 1975 · by jerry gay

field notes: the first bug

sept 9, 1947. operators of the Harvard Mark II computer found an actual moth stuck in a relay. so they taped it into the logbook.



engineers had already been calling glitches "bugs" for decades. but this one had wings.

the Mandelbug

Bugus fractalis



so tangled its cause looks like pure chaos. zoom in: more bug. zoom in again: still bug. it's bugs all the way down.

habitat: systems with 40 microservices

catch difficulty: ★★★★★

a pocket field guide to

WILD ERRORS

how to spot & identify the numbers that pop up when the web has something to tell you

field guides to the digital wild · no. 2

my sightings

- ☐ 200 (so common nobody looks twice)
- ☐ 301 ☐ 403 ☐ 503
- ☐ 404 (if you spotted one, you didn't.)
- ☐ 418 (rare!)
- ☐ _____ (go find more.)

if you meet one:
4xx: check what you asked for, and whether you're allowed.
5xx: it's not you. wait, then try again.

made with ink & a slow connection
after RFC 9110, 2022 & RFC 2324, 1998
by jerry gay

cut along the dashes

field notes: the rarest one

april 1, 1998. someone wrote a serious-looking standard for brewing coffee over the internet. ask a teapot for coffee and it answers:

418, i'm a teapot.

in 2017, engineers tried to remove it. people protested until it stayed. today it's officially reserved: a protected species.

the 503

Servus hibernans

too tired, too busy, or out for maintenance. leave it alone and it's usually back by morning. (poking it makes it grumpier.)

habitat: launch days, the minute tickets go on sale

catch difficulty: ★★☆☆☆
(just wait)

the 403

Clientus prohibitus

guards the gate. its cousin the 401 asks "who are you?" the 403 knows exactly who you are. the answer is still no.

habitat: admin pages, the cookie jar

catch difficulty: don't. it's doing its job.

the 404

Clientus absens

you've seen its sign a hundred times. but go looking for the page and you only find the spot where it should be. (it isn't hiding. there's just nothing there.)

habitat: links from 2004, typos

catch difficulty: ☆☆☆☆☆
(nothing to catch)

the 301

Redirectus migratorius

flew away and never came back. but it's polite: it leaves a note saying where it went, and your browser follows without a word.

habitat: old addresses, sites that moved to https

catch difficulty: ★☆☆☆☆
(it tells you where it went)

how to identify

every one of them carries a number. look at the first digit: it tells you the family.

2 happy. it worked.
3 migratory. it moved.
4 the problem's on your side.
5 the server tripped.

(1xx are the larvae. you'll almost never see one.)